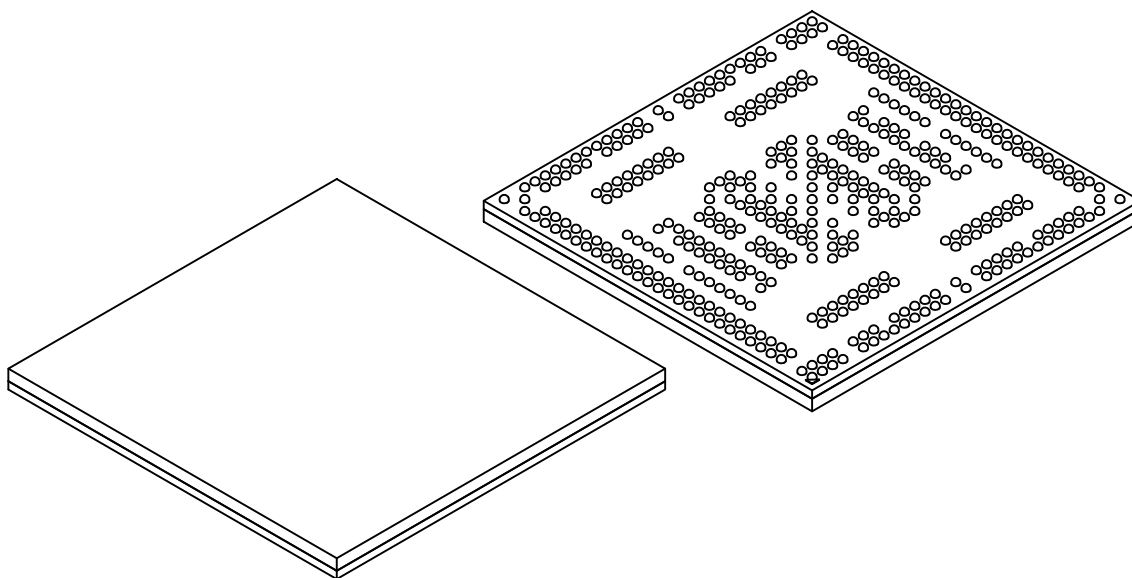


Packaging Diagrams and Parameters

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	442		
Pitch	e	0.50 BSC		
Overall Height	A	—	—	1.20
Ball Height	A1	0.16	0.21	0.26
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.35 REF		
Overall Length	D	16.00 BSC		
Ball Array Length	D2	15.00 BSC		
Overall Width	E	16.00 BSC		
Ball Array Width	E2	15.00 BSC		
Ball Diameter	b	0.27	0.32	0.37

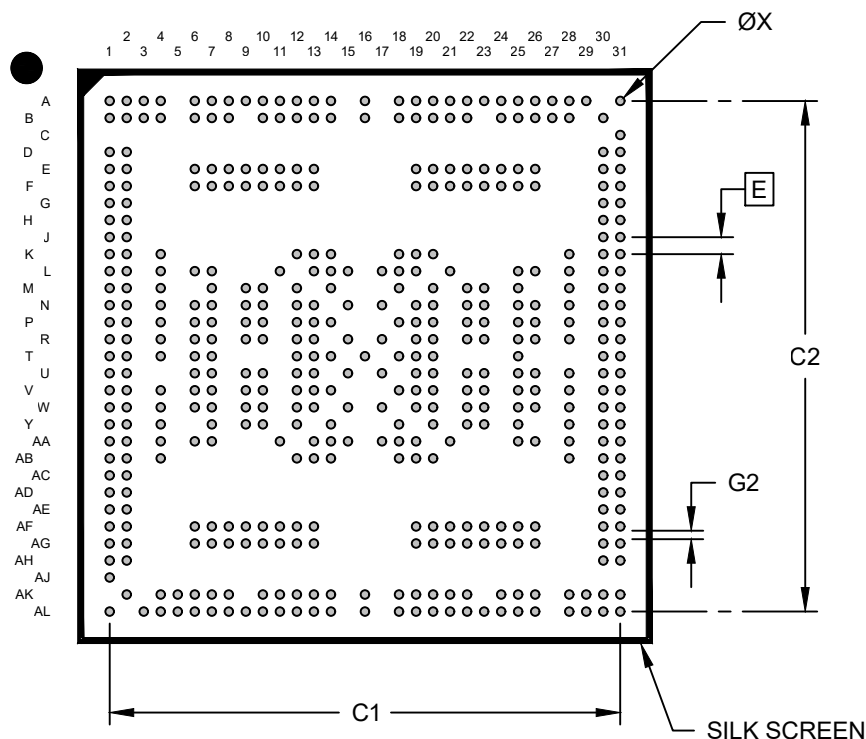
Notes:

1. The A1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
REF: Reference Dimension, usually without tolerance, is for information purposes only.

Land Pattern

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		15.00	
Contact Pad Spacing	C2		15.00	
Contact Pad Diameter (X442)	X			0.25
Contact Pad to Contact Pad	G2	0.25		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, please refer to the current industry standard IPC-7093.